

Learning Objectives

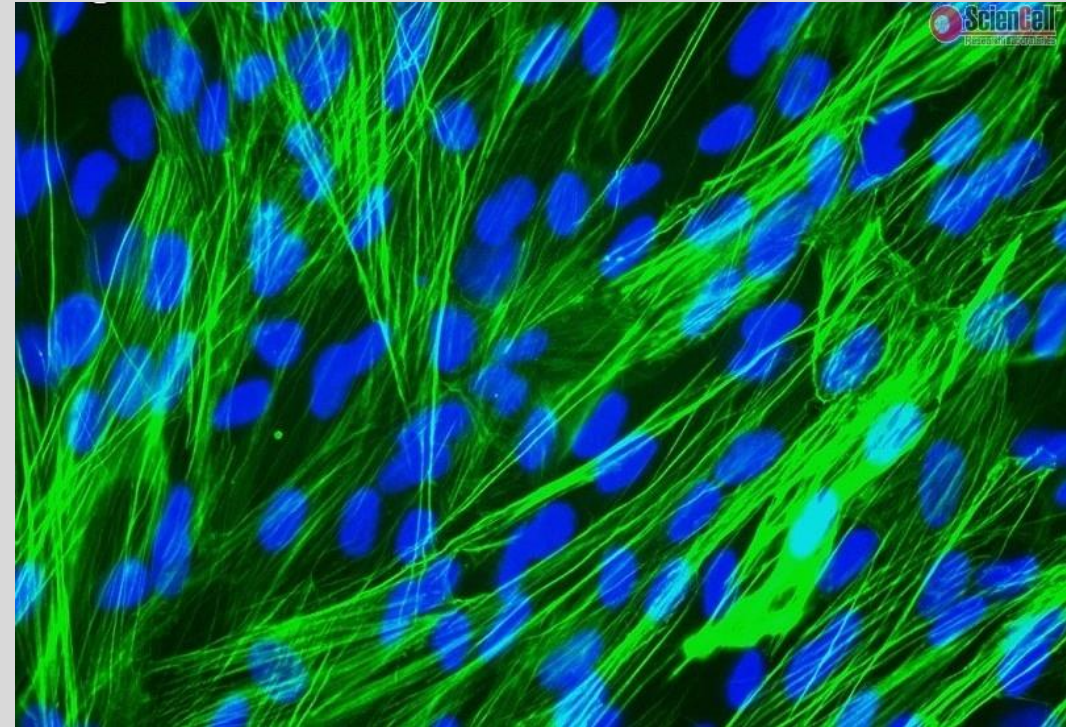
- Learning Objective 1: Describe the main mechanisms by which VSMC contracts and relaxes
- Learning Objective 2: Identify contractile proteins from VSMC and describe the source and movement of Ca^{2+} ions in the vascular smooth muscle cells.
- Learning Objective 3: Explain the mechanisms by which KCl induces vascular contraction
- Learning Objective 4: Recognize major phenotypic changes in diseased VSMC

Vascular Smooth Muscle Cells (VSMC)

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Spindle-shaped cells with singles nucleus

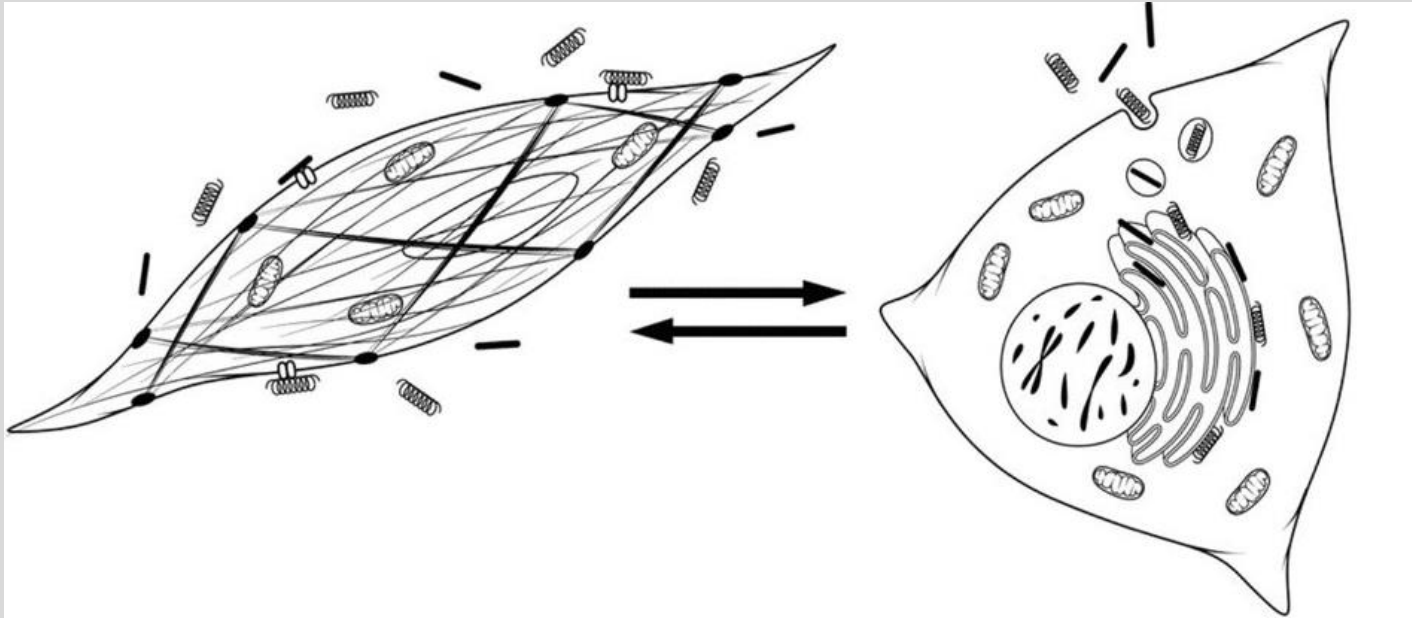


Images from Sepulveda's Laboratory- NYITCOM

Normal and Diseased VSMC

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Normal

Spindle shape

Contractile

Nonproliferating

Nonmigrating

Diseased

Rhomboid shape

Secretory/Synthetic (synthesizing
extracellular matrix)

Proliferating (cell division)

Migrating (able to invade)

Practice Question

A 59-year-old man becomes unconscious after experiencing severe retrosternal chest pain. He has a history of smoking and dyslipidemia but is noncompliant with his prescribed medication regimen. The man is pronounced dead on arrival to the emergency room. An autopsy reveals diffuse atherosclerotic vascular disease involving the coronary arteries. One of the atheromatous plaques in the left main artery has a ruptured fibrous cap with an overlying occluding thrombus. Which of the following cellular components is most directly involved in formation of the fibrous portion of this patient's plaque?

- a) Endothelial Cells
- b) Activated Platelets
- c) Smooth Muscle Cells
- d) Macrophages
- e) Infiltrating T lymphocytes

Structure of VSMC

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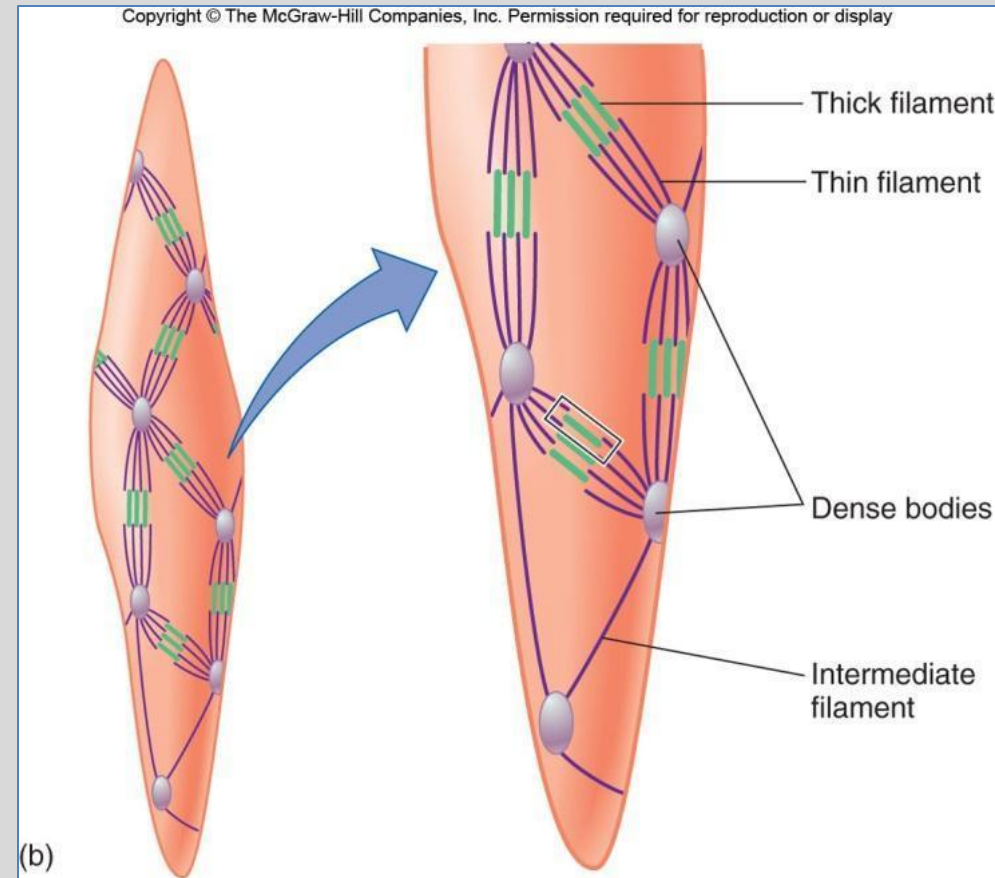
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Cytoskeletal filaments

1. Thin filaments : Actin (**purple**)
2. Thick filaments: Myosin (**green**)
3. Intermediate filaments: Desmin and Vimentin. Support the shape of the smooth muscle cells and hold the dense bodies in place

Dense bodies

Located in the cytoplasm and at the cell membrane.

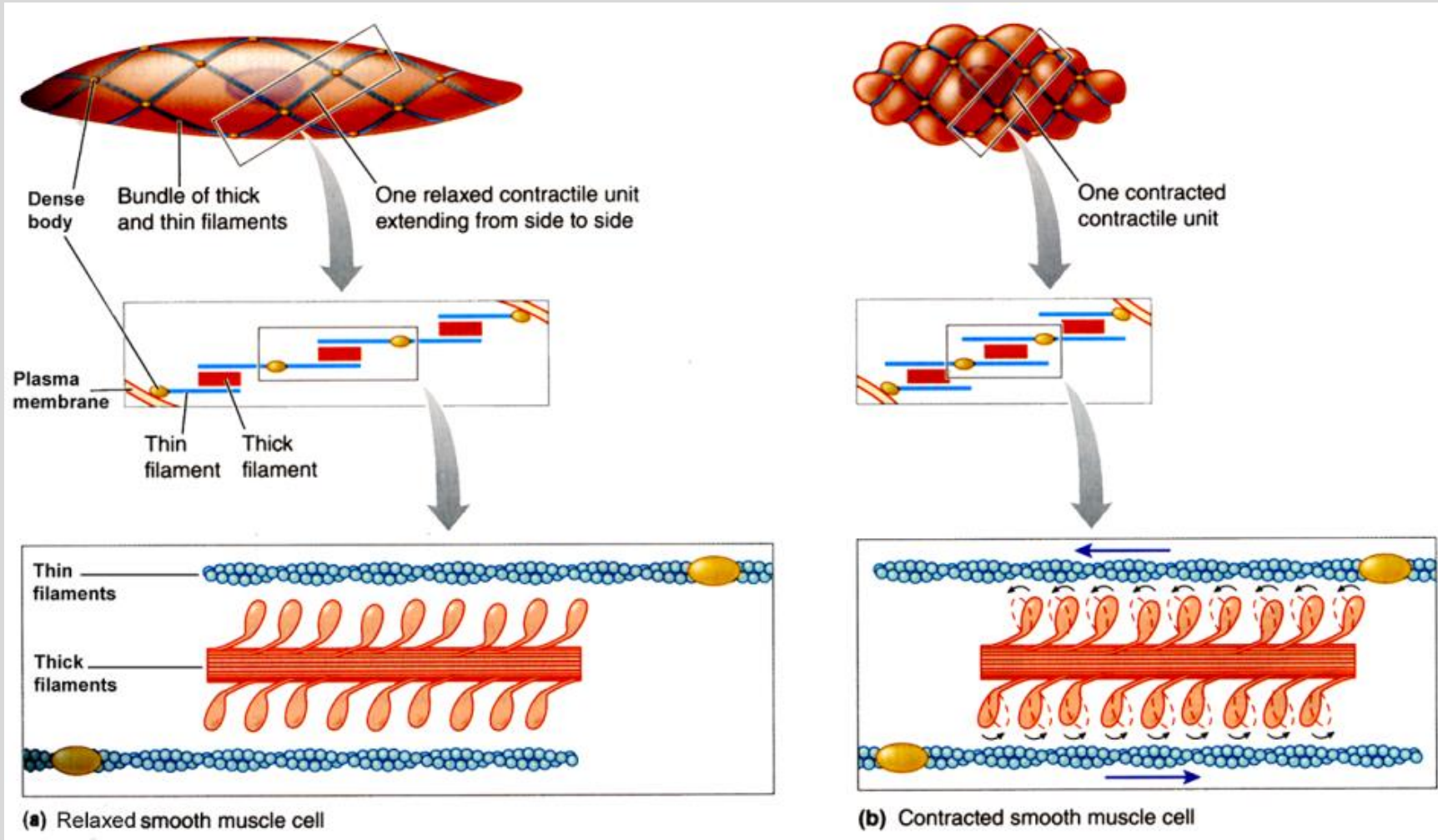


Sherwood, Human Physiology: From Cells to Systems, 8th ed., Brooks/Cole, 2013
(Chapter 8: pp. 294-295)

Arrangement of Contractile Proteins

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Neural Control of VSMC Contraction

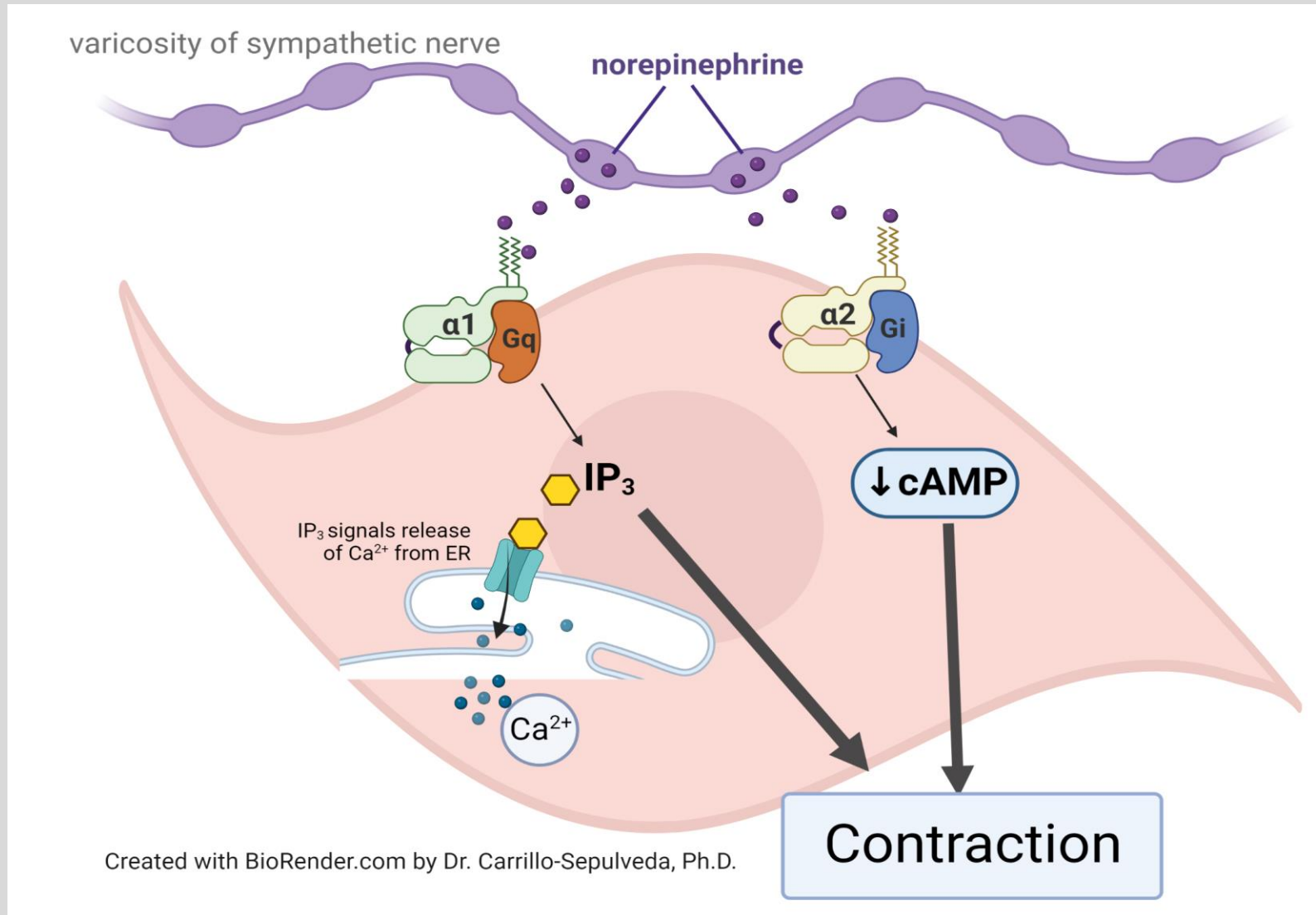
Neurotransmitter released by:

- Sympathetic Fiber: Norepinephrine. Alpha 1 and Alpha 2 adrenergic receptors

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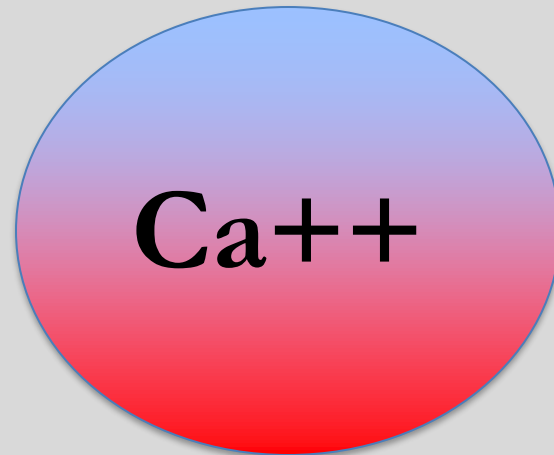
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SR instead of ER



Contraction

After increases of calcium cytosolic, what is the next step of vascular smooth muscle contraction?

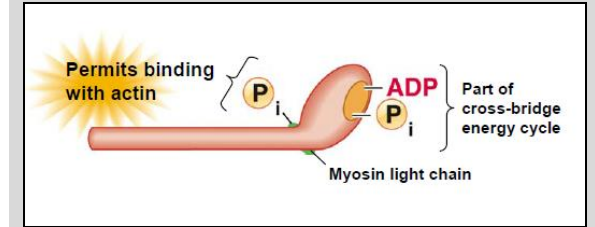
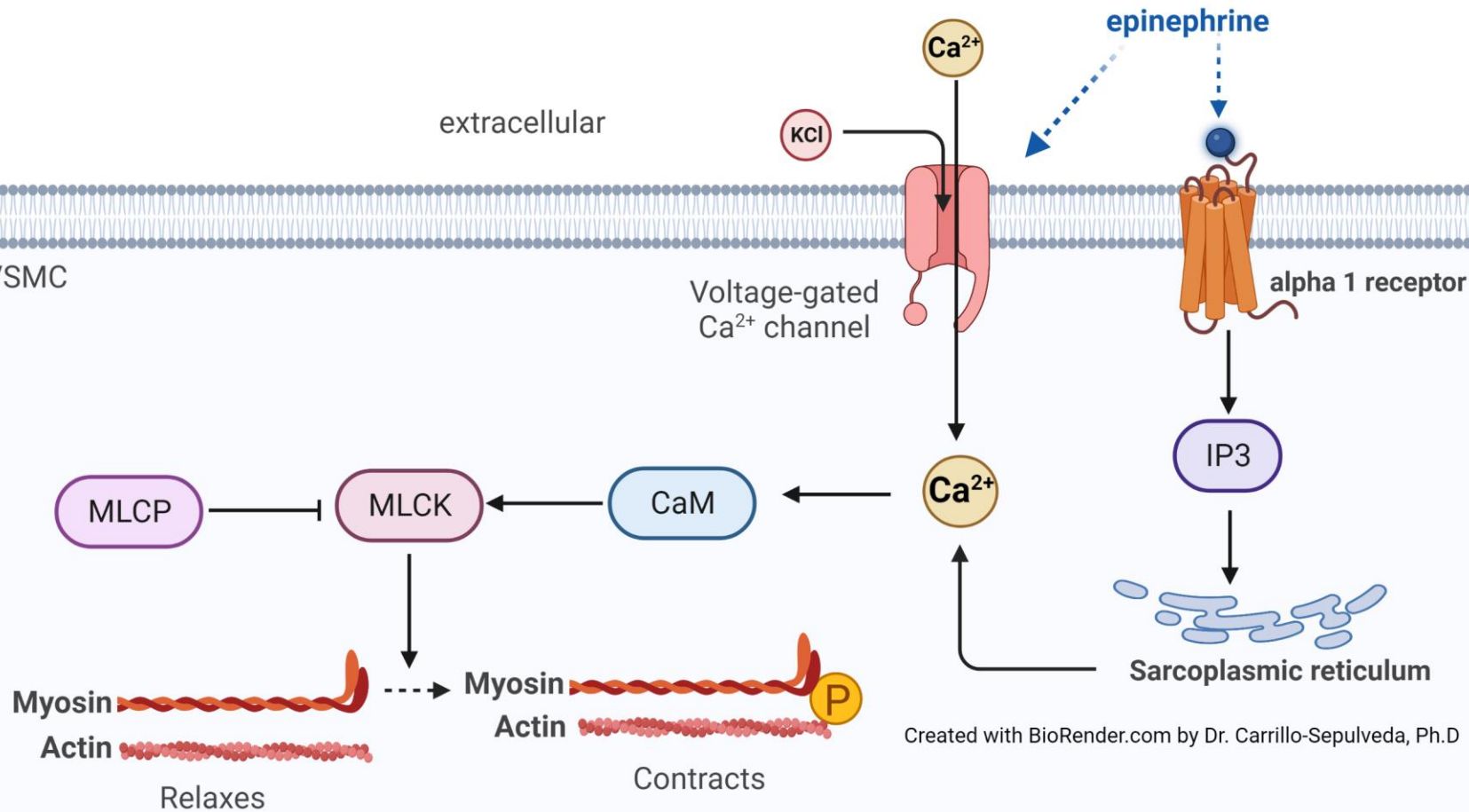


Vascular Smooth Muscle Contraction

Chemical event

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Unlike myosin light chain kinase, which is only active when cytoplasmic Ca^{2+} is elevated, *myosin light chain phosphatase is **always** active.*

Mechanism of VSMC Contraction

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↑Cytosolic Ca^{++}



Ca^{++} + Calmodulin



Activation of myosin kinase



Phosphorylation of myosin light chain



Attachment of myosin head with actin



Contraction by sliding filament mechanism



Contraction continues as
long as myosin is
phosphorylated

Practice Question

You are working in a scientific project which aim to develop a new vasodilator that can replace nitroglycerine. This new drug will be used to treat elevated blood pressure (hypertensive emergency). Which of the following intracellular events is most likely responsible for the beneficial effects of this “new vasodilator”?

- a. Actin phosphorylation
- b. Calcium release from sarcoplasmic reticulum
- c. Tyrosine kinase activation
- d. Myosin dephosphorylation

Ca⁺⁺ Source in VSMC

- Smooth muscle uses Ca⁺⁺ from both extracellular fluid and the sarcoplasmic reticulum

Source of extracellular Ca²⁺ ions :

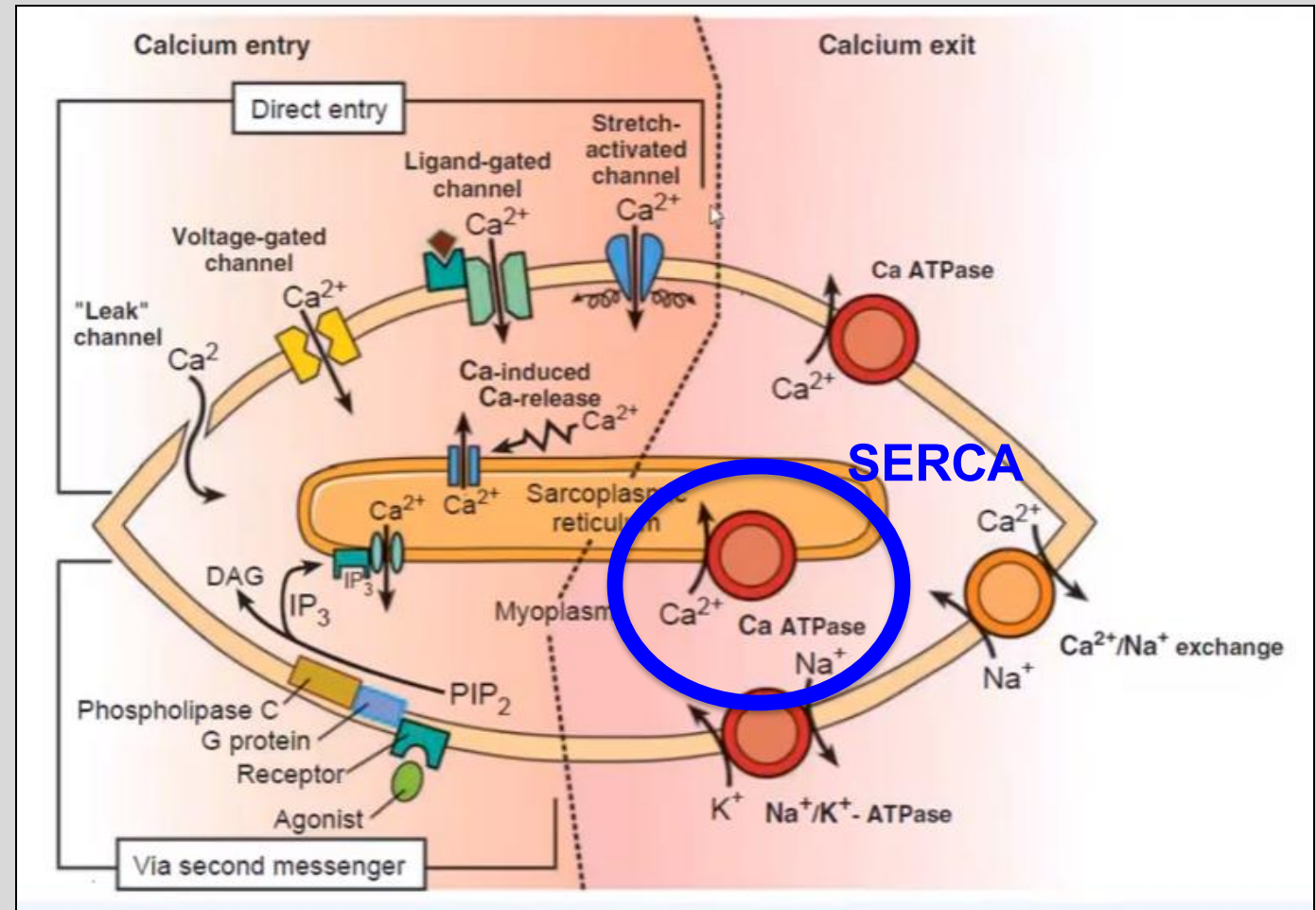
- Voltage-gated L-type Ca²⁺ channels
- Voltage-independent Ca²⁺ channels (ligand-gated, stretch activated channel, and “leak” channels)

Source of intracellular Ca²⁺ ions:

- Ca²⁺-induced Ca²⁺-release mechanism
- Inositol 1,4,5-triphosphate (IP₃)-mediated Ca²⁺ release

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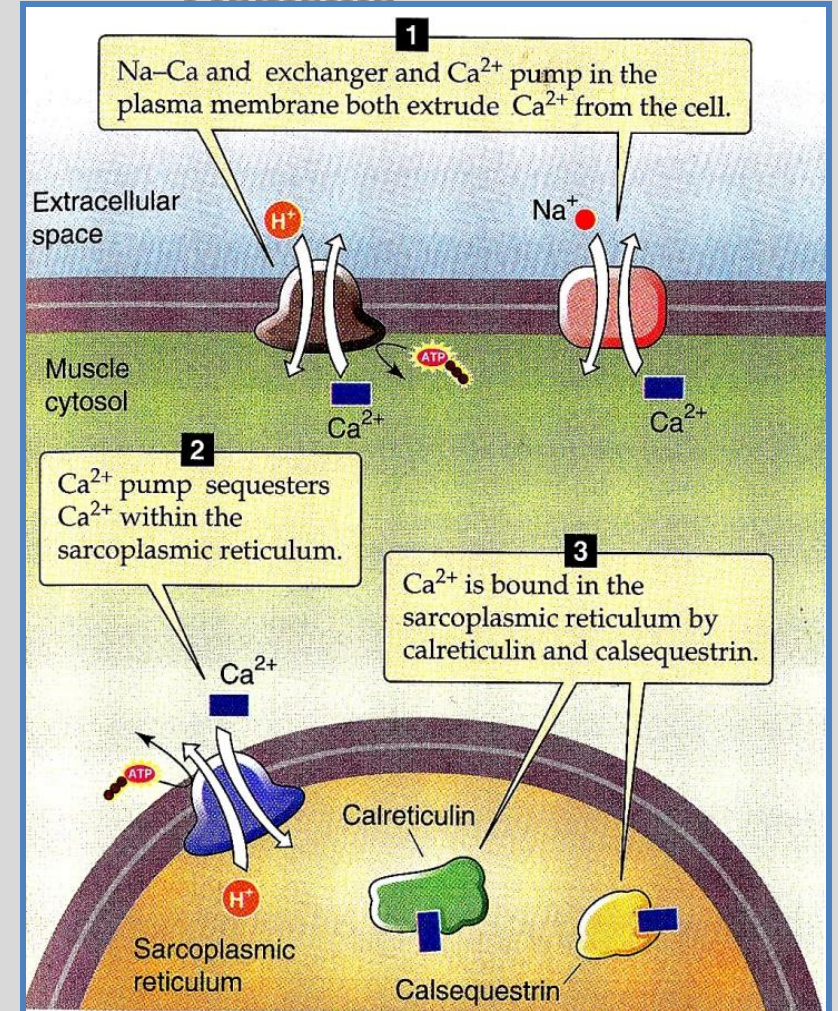
Vascular Smooth Muscle Relaxation

After a contraction-activating stimulus (electrical or chemical) has subsided, *Ca²⁺ ions must be removed from the cytoplasm for muscle relaxation to occur.*

- Extracellular extrusion of Ca²⁺ by Ca²⁺ pump and Na⁺/Ca²⁺ exchanger
- Ca²⁺ re-uptake in sarcoplasmic reticulum by Ca²⁺-ATPase (SERCA)
- Intracellular Ca²⁺ concentration falls to resting level
- Formation of Ca²⁺-calmodulin complexes decreases
- MLCK activity decreases
- MLC phosphatase dephosphorylates MLC on myosin heads

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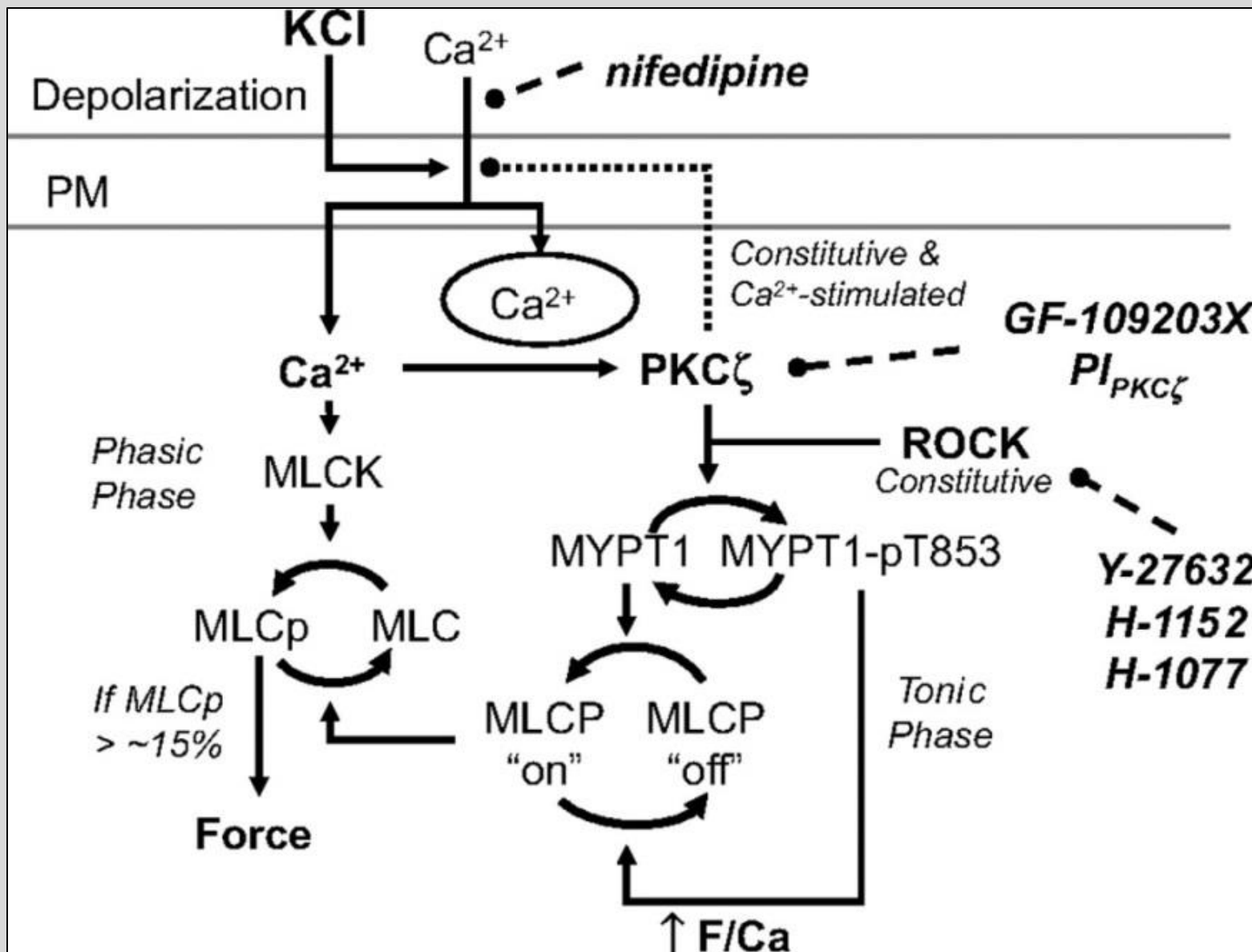
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KCl causes VSMC contraction

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- Nifedipine
 - Amlodipine
 - Verapamil
- All L-type calcium blocker

Summary

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- Contractile proteins involved in the VSMC contraction
- Mechanism of VSMC contraction: calcium plus calmodulin complex
- Effects of sodium nitroprusside and KCL in VSMC